

Coding Challenge #3- SQL Crime Management Schema

```
CREATE database Crime_Management;
```

```
USE Crime_Management;
```

```
CREATE TABLE Crime (  
    CrimeID INT PRIMARY KEY,  
    IncidentType VARCHAR(255),  
    IncidentDate DATE,  
    Location VARCHAR(255),  
    Description TEXT,  
    Status VARCHAR(20)  
);
```

```
CREATE TABLE Victim (  
    VictimID INT PRIMARY KEY,  
    CrimeID INT,  
    Name VARCHAR(255),  
    ContactInfo VARCHAR(255),  
    Injuries VARCHAR(255),  
    FOREIGN KEY (CrimeID) REFERENCES Crime(CrimeID)  
);
```

```
CREATE TABLE Suspect (  
    SuspectID INT PRIMARY KEY,
```

```
CrimeID INT,  
Name VARCHAR(255),  
Description TEXT,  
CriminalHistory TEXT,  
FOREIGN KEY (CrimeID) REFERENCES Crime(CrimeID)  
);
```

```
INSERT INTO Crime (CrimeID, IncidentType, IncidentDate, Location, Description,  
Status)
```

```
VALUES
```

```
(1, 'Robbery', '2023-09-15', '123 Main St, Cityville', 'Armed robbery at a  
convenience store', 'Open'),
```

```
(2, 'Homicide', '2023-09-20', '456 Elm St, Townsville', 'Investigation into a  
murder case', 'Under  
Investigation'),
```

```
(3, 'Theft', '2023-09-10', '789 Oak St, Villagetown', 'Shoplifting incident at a  
mall', 'Closed');
```

```
INSERT INTO Victim (VictimID, CrimeID, Name, ContactInfo, Injuries)
```

```
VALUES
```

```
(1, 1, 'John Doe', 'johndoe@example.com', 'Minor injuries'),
```

```
(2, 2, 'Jane Smith', 'janesmith@example.com', 'Deceased'),
```

```
(3, 3, 'Alice Johnson', 'alicejohnson@example.com', 'None');
```

```
INSERT INTO Suspect (SuspectID, CrimeID, Name, Description, CriminalHistory)
```

```
VALUES
```

```
(1, 1, 'Robber 1', 'Armed and masked robber', 'Previous robbery convictions'),
```

(2, 2, 'Unknown', 'Investigation ongoing', NULL),

(3, 3, 'Suspect 1', 'Shoplifting suspect', 'Prior shoplifting arrests');

1. Select all open incidents.

SELECT * FROM Crime WHERE Status = 'Open';

The screenshot shows a database application interface. At the top, a SQL query is entered: `SELECT * FROM Crime WHERE Status = 'Open';`. Below the query, a 'Result Grid' displays the results of the query. The grid has columns: CrimeID, IncidentType, IncidentDate, Location, Description, and Status. The first row shows a robbery incident that is open.

CrimeID	IncidentType	IncidentDate	Location	Description	Status
1	Robbery	2023-09-15	123 Main St, Cityville	Armed robbery at a convenience store	Open

Below the result grid, an 'Action Output' section shows a log of database actions and their results, including several SELECT queries and their return counts.

2. Find the total number of incidents.

SELECT COUNT(*) AS TotalIncidents FROM Crime;

The screenshot shows the same database application interface. A new SQL query is entered: `SELECT COUNT(*) AS TotalIncidents FROM Crime;`. The 'Result Grid' shows a single row with the value 3 in the 'TotalIncidents' column. The 'Action Output' section at the bottom shows a log of database actions, including the current query and its result.

TotalIncidents
3

3. List all unique incident types.

SELECT DISTINCT IncidentType FROM Crime;

```

55 #3
56 • SELECT DISTINCT IncidentType FROM Crime;
57

```

Result Grid

IncidentType
Robbery
Homicide
Theft

Crime 24 × Read Only

Output

Action Output

#	Time	Action	Message
32	10:30:02	SELECT * FROM Crime WHERE IncidentType = 'Homicide' AND EXISTS (SELECT 1 FROM Crime c2 ...	1 row(s) returned
33	10:30:28	SELECT c.CrimeID, c.IncidentType, COALESCE(s.Name, 'No Suspect') AS SuspectName FROM Crime c LEF...	3 row(s) returned
34	10:30:50	SELECT s.* FROM Suspect s JOIN Crime c ON s.CrimeID = c.CrimeID WHERE c.IncidentType IN ('Robbery', ...	1 row(s) returned
35	12:50:11	SELECT * FROM Crime WHERE Status = 'Open' LIMIT 0, 1000	1 row(s) returned
36	12:52:43	SELECT COUNT(*) AS TotalIncidents FROM Crime LIMIT 0, 1000	1 row(s) returned
37	12:55:11	SELECT DISTINCT IncidentType FROM Crime LIMIT 0, 1000	3 row(s) returned

4. Retrieve incidents that occurred between '2023-09-01' and '2023-09-10'.

`SELECT * FROM Crime`

`WHERE IncidentDate BETWEEN '2023-09-01' AND '2023-09-10';`

```

58 #4
59 • SELECT * FROM Crime
60 WHERE IncidentDate BETWEEN '2023-09-01' AND '2023-09-10';

```

Result Grid

CrimeID	IncidentType	IncidentDate	Location	Description	Status
3	Theft	2023-09-10	789 Oak St, Villagetown	Shoplifting incident at a mall	Closed

Crime 25 × Apply Revert

Output

Action Output

#	Time	Action	Message
33	10:30:28	SELECT c.CrimeID, c.IncidentType, COALESCE(s.Name, 'No Suspect') AS SuspectName FROM Crime c LEF...	3 row(s) returned
34	10:30:50	SELECT s.* FROM Suspect s JOIN Crime c ON s.CrimeID = c.CrimeID WHERE c.IncidentType IN ('Robbery', ...	1 row(s) returned
35	12:50:11	SELECT * FROM Crime WHERE Status = 'Open' LIMIT 0, 1000	1 row(s) returned
36	12:52:43	SELECT COUNT(*) AS TotalIncidents FROM Crime LIMIT 0, 1000	1 row(s) returned
37	12:55:11	SELECT DISTINCT IncidentType FROM Crime LIMIT 0, 1000	3 row(s) returned
38	13:00:26	SELECT * FROM Crime WHERE IncidentDate BETWEEN '2023-09-01' AND '2023-09-10' LIMIT 0, 1000	1 row(s) returned

5. List persons involved in incidents in descending order of age.

`ALTER TABLE Victim ADD Age INT;`

`UPDATE Victim`

```

SET Age = CASE VictimID
    WHEN 1 THEN 34 -- John Doe
    WHEN 2 THEN 29 -- Jane Smith
    WHEN 3 THEN 22 -- Alice Johnson
END;
ALTER TABLE Suspect ADD Age INT;

```

```

UPDATE Suspect
SET Age = CASE SuspectID
    WHEN 1 THEN 40 -- Robber 1
    WHEN 2 THEN NULL -- Unknown
    WHEN 3 THEN 30 -- Suspect 1
END;

```

```

SELECT Name, Age FROM Victim
UNION
SELECT Name, Age FROM Suspect
ORDER BY Age DESC;

```

The screenshot shows a database management tool interface. At the top, a SQL query is entered in a text area:

```

79 #5
80 • SELECT Name, Age FROM Victim
81 UNION
82 SELECT Name, Age FROM Suspect
83 ORDER BY Age DESC;
84

```

Below the query editor, the 'Result Grid' is displayed, showing the results of the query. The grid has two columns: 'Name' and 'Age'. The results are as follows:

Name	Age
Robber 1	40
John Doe	34
Suspect 1	30
Jane Smith	29
Alice Johnson	22
Unknown	NULL

At the bottom of the screenshot, the 'Output' pane shows a list of actions and their results. The last action is the query executed, which returned 6 rows.

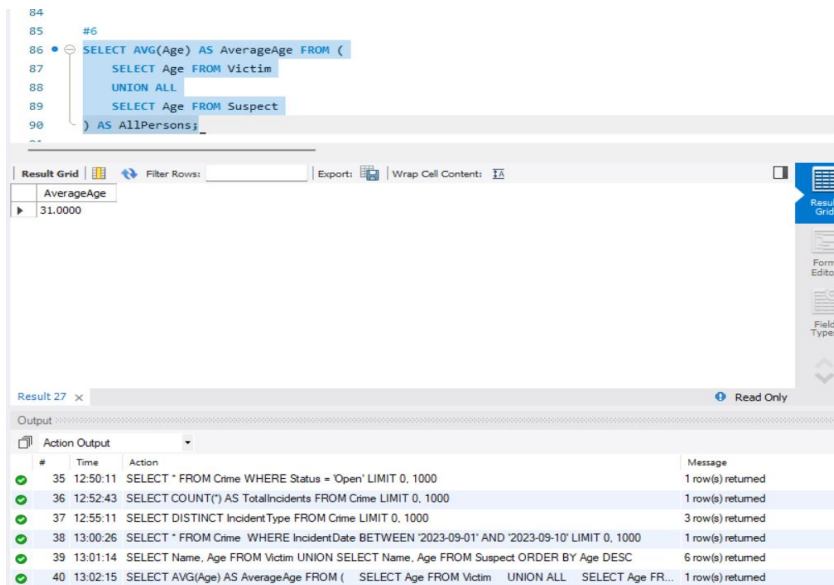
#	Time	Action	Message
34	10:30:50	SELECT s.* FROM Suspect s JOIN Crime c ON s.CrimeID = c.CrimeID WHERE c.IncidentType IN ('Robbery', ...	1 row(s) returned
35	12:50:11	SELECT * FROM Crime WHERE Status = 'Open' LIMIT 0, 1000	1 row(s) returned
36	12:52:43	SELECT COUNT(*) AS TotalIncidents FROM Crime LIMIT 0, 1000	1 row(s) returned
37	12:55:11	SELECT DISTINCT IncidentType FROM Crime LIMIT 0, 1000	3 row(s) returned
38	13:00:26	SELECT * FROM Crime WHERE IncidentDate BETWEEN '2023-09-01' AND '2023-09-10' LIMIT 0, 1000	1 row(s) returned
39	13:01:14	SELECT Name, Age FROM Victim UNION SELECT Name, Age FROM Suspect ORDER BY Age DESC	6 row(s) returned

6. Find the average age of persons involved in incidents.

```

SELECT AVG(Age) AS AverageAge FROM (
    SELECT Age FROM Victim
    UNION ALL
    SELECT Age FROM Suspect
) AS AllPersons;

```



7. List incident types and their counts, only for open cases.

```

SELECT IncidentType, COUNT(*) AS OpenCaseCount
FROM Crime
WHERE Status = 'Open'
GROUP BY IncidentType;

```

```

92  #7
93  SELECT IncidentType, COUNT(*) AS OpenCaseCount
94  FROM Crime
95  WHERE Status = 'Open'
96  GROUP BY IncidentType;
97
98  #8

```

IncidentType	OpenCaseCount
Robbery	1

Result 28 x Read Only

Output

#	Time	Action	Message
36	12:52:43	SELECT COUNT(*) AS TotalIncidents FROM Crime LIMIT 0, 1000	1 row(s) returned
37	12:55:11	SELECT DISTINCT IncidentType FROM Crime LIMIT 0, 1000	3 row(s) returned
38	13:00:26	SELECT * FROM Crime WHERE IncidentDate BETWEEN '2023-09-01' AND '2023-09-10' LIMIT 0, 1000	1 row(s) returned
39	13:01:14	SELECT Name, Age FROM Victim UNION SELECT Name, Age FROM Suspect ORDER BY Age DESC	6 row(s) returned
40	13:02:15	SELECT AVG(Age) AS AverageAge FROM (SELECT Age FROM Victim UNION ALL SELECT Age FR...	1 row(s) returned
41	13:03:35	SELECT IncidentType, COUNT(*) AS OpenCaseCount FROM Crime WHERE Status = 'Open' GROUP BY In...	1 row(s) returned

8. Find persons with names containing 'Doe'.

SELECT Name FROM Victim WHERE Name LIKE '%Doe%'

UNION

SELECT Name FROM Suspect WHERE Name LIKE '%Doe%';

```

98  #8
99  SELECT Name FROM Victim WHERE Name LIKE '%Doe%'
100 UNION
101 SELECT Name FROM Suspect WHERE Name LIKE '%Doe%';
102
103 #9
104 -- Open cases

```

Name
John Doe

Result 29 x Read Only

Output

#	Time	Action	Message
37	12:55:11	SELECT DISTINCT IncidentType FROM Crime LIMIT 0, 1000	3 row(s) returned
38	13:00:26	SELECT * FROM Crime WHERE IncidentDate BETWEEN '2023-09-01' AND '2023-09-10' LIMIT 0, 1000	1 row(s) returned
39	13:01:14	SELECT Name, Age FROM Victim UNION SELECT Name, Age FROM Suspect ORDER BY Age DESC	6 row(s) returned
40	13:02:15	SELECT AVG(Age) AS AverageAge FROM (SELECT Age FROM Victim UNION ALL SELECT Age FR...	1 row(s) returned
41	13:03:35	SELECT IncidentType, COUNT(*) AS OpenCaseCount FROM Crime WHERE Status = 'Open' GROUP BY In...	1 row(s) returned
42	13:04:26	SELECT Name FROM Victim WHERE Name LIKE '%Doe%'; UNION SELECT Name FROM Suspect WHERE ...	1 row(s) returned

9. Retrieve the names of persons involved in open cases and closed cases.

-- Open cases

SELECT Name FROM Victim

WHERE CrimeID IN (SELECT CrimeID FROM Crime WHERE Status = 'Open')

UNION

SELECT Name FROM Suspect

WHERE CrimeID IN (SELECT CrimeID FROM Crime WHERE Status = 'Open');

-- Closed cases

SELECT Name FROM Victim

WHERE CrimeID IN (SELECT CrimeID FROM Crime WHERE Status = 'Closed')

UNION

SELECT Name FROM Suspect

WHERE CrimeID IN (SELECT CrimeID FROM Crime WHERE Status = 'Closed');

The screenshot shows a database management tool interface. At the top, a SQL query is displayed in a text editor. The query is a UNION of two SELECT statements. The first SELECT statement retrieves the Name from the Victim table where the CrimeID is in a subquery of CrimeID from the Crime table where Status is 'Open'. The second SELECT statement retrieves the Name from the Suspect table where the CrimeID is in a subquery of CrimeID from the Crime table where Status is 'Open'. Below the query editor, the 'Result Grid' is visible, showing a table with two columns: 'Name' and 'Suspect 1'. The 'Name' column contains the value 'Alice Johnson'. To the right of the 'Result Grid' is a sidebar with icons for 'Result Grid', 'Form Editor', and 'Field Types'. At the bottom of the screenshot, the 'Output' pane shows a log of actions and their results. The log includes the following entries:

#	Time	Action	Message
38	13:00:26	SELECT * FROM Crime WHERE IncidentDate BETWEEN '2023-09-01' AND '2023-09-10' LIMIT 0, 1000	1 row(s) returned
39	13:01:14	SELECT Name, Age FROM Victim UNION SELECT Name, Age FROM Suspect ORDER BY Age DESC	6 row(s) returned
40	13:02:15	SELECT AVG(Age) AS AverageAge FROM (SELECT Age FROM Victim UNION ALL SELECT Age FR...	1 row(s) returned
41	13:03:35	SELECT IncidentType, COUNT(*) AS OpenCaseCount FROM Crime WHERE Status = 'Open' GROUP BY In...	1 row(s) returned
42	13:04:26	SELECT Name FROM Victim WHERE Name LIKE '%Doe%'; UNION SELECT Name FROM Suspect WHERE ...	1 row(s) returned
43	13:05:38	SELECT Name FROM Victim WHERE CrimeID IN (SELECT CrimeID FROM Crime WHERE Status = 'Closed') ...	2 row(s) returned

10. List incident types where there are persons aged 30 or 35 involved.

SELECT DISTINCT c.IncidentType

FROM Crime c

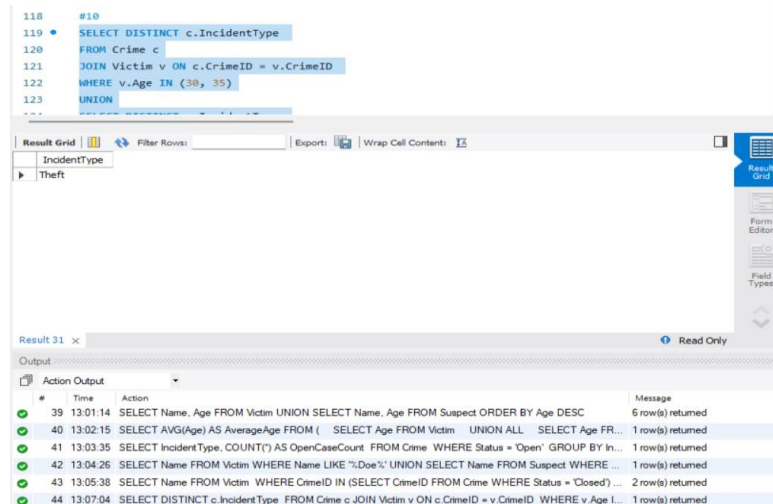
JOIN Victim v ON c.CrimeID = v.CrimeID

WHERE v.Age IN (30, 35)


```

UNION
SELECT DISTINCT c.IncidentType
FROM Crime c
JOIN Suspect s ON c.CrimeID = s.CrimeID
WHERE s.Age IN (30, 35);

```



11. Find persons involved in incidents of the same type as 'Robbery'.

```

SELECT Name FROM Victim
WHERE CrimeID IN (
    SELECT CrimeID FROM Crime WHERE IncidentType = 'Robbery'
)
UNION
SELECT Name FROM Suspect
WHERE CrimeID IN (
    SELECT CrimeID FROM Crime WHERE IncidentType = 'Robbery'
);

```

129 #11
 130 • SELECT Name FROM Victim
 131 WHERE CrimeID IN (
 132 SELECT CrimeID FROM Crime WHERE IncidentType = 'Robbery'
 133)
 134 UNION
 135 SELECT Name FROM Suspect

Result Grid

Name
John Doe
Robber 1

Result 32 x Read Only

Output

Action Output

#	Time	Action	Message
40	13:02:15	SELECT AVG(Age) AS AverageAge FROM (SELECT Age FROM Victim UNION ALL SELECT Age FR...	1 row(s) returned
41	13:03:35	SELECT IncidentType, COUNT(*) AS OpenCaseCount FROM Crime WHERE Status = 'Open' GROUP BY In...	1 row(s) returned
42	13:04:26	SELECT Name FROM Victim WHERE Name LIKE '%Doe%' UNION SELECT Name FROM Suspect WHERE ...	1 row(s) returned
43	13:05:38	SELECT Name FROM Victim WHERE CrimeID IN (SELECT CrimeID FROM Crime WHERE Status = 'Closed') ...	2 row(s) returned
44	13:07:04	SELECT DISTINCT c.IncidentType FROM Crime c JOIN Victim v ON c.CrimeID = v.CrimeID WHERE v.Age I...	1 row(s) returned
45	13:07:57	SELECT Name FROM Victim WHERE CrimeID IN (SELECT CrimeID FROM Crime WHERE IncidentType =...	2 row(s) returned

12. List incident types with more than one open case.

SELECT IncidentType, COUNT(*) AS OpenCount

FROM Crime

WHERE Status = 'Open'

GROUP BY IncidentType

HAVING COUNT(*) > 1;

140 #12
 141 • SELECT IncidentType, COUNT(*) AS OpenCount
 142 FROM Crime
 143 WHERE Status = 'Open'
 144 GROUP BY IncidentType
 145 HAVING COUNT(*) > 1;
 146

Result Grid

IncidentType	OpenCount
--------------	-----------

Result 33 x Read Only

Output

Action Output

#	Time	Action	Message
41	13:03:35	SELECT IncidentType, COUNT(*) AS OpenCaseCount FROM Crime WHERE Status = 'Open' GROUP BY In...	1 row(s) returned
42	13:04:26	SELECT Name FROM Victim WHERE Name LIKE '%Doe%' UNION SELECT Name FROM Suspect WHERE ...	1 row(s) returned
43	13:05:38	SELECT Name FROM Victim WHERE CrimeID IN (SELECT CrimeID FROM Crime WHERE Status = 'Closed') ...	2 row(s) returned
44	13:07:04	SELECT DISTINCT c.IncidentType FROM Crime c JOIN Victim v ON c.CrimeID = v.CrimeID WHERE v.Age I...	1 row(s) returned
45	13:07:57	SELECT Name FROM Victim WHERE CrimeID IN (SELECT CrimeID FROM Crime WHERE IncidentType =...	2 row(s) returned
46	13:09:05	SELECT IncidentType, COUNT(*) AS OpenCount FROM Crime WHERE Status = 'Open' GROUP BY Inciden...	0 row(s) returned

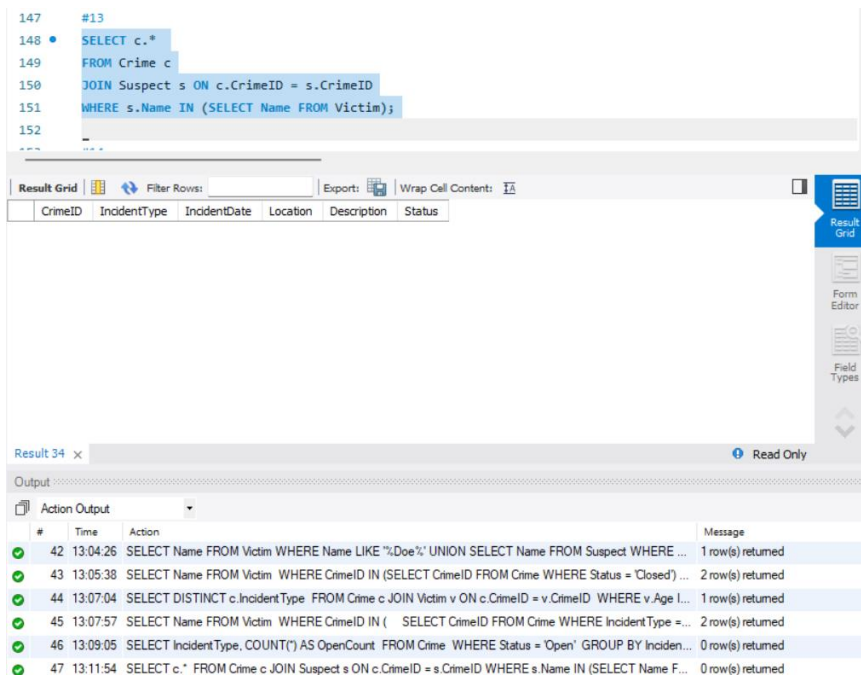
13. List all incidents with suspects whose names also appear as victims in other incidents.

SELECT c.*

FROM Crime c

JOIN Suspect s ON c.CrimeID = s.CrimeID

WHERE s.Name IN (SELECT Name FROM Victim);



14. Retrieve all incidents along with victim and suspect details.

SELECT c.*, v.Name AS VictimName, s.Name AS SuspectName

FROM Crime c

LEFT JOIN Victim v ON c.CrimeID = v.CrimeID

LEFT JOIN Suspect s ON c.CrimeID = s.CrimeID;

```

152
153 #14
154 • SELECT c.*, v.Name AS VictimName, s.Name AS SuspectName
155 FROM Crime c
156 LEFT JOIN Victim v ON c.CrimeID = v.CrimeID
157 LEFT JOIN Suspect s ON c.CrimeID = s.CrimeID;
158

```

CrimeID	IncidentType	IncidentDate	Location	Description	Status	VictimName	SuspectName
1	Robbery	2023-09-15	123 Main St, Cityville	Armed robbery at a convenience store	Open	John Doe	Robert Johnson
2	Homicide	2023-09-20	456 Elm St, Townsville	Investigation into a murder case	Under Investigation	Jane Smith	Unknown
3	Theft	2023-09-10	789 Oak St, Villagetown	Shoplifting incident at a mall	Closed	Alice Johnson	Suspect

Result 35 x

Output

#	Time	Action	Message
43	13:05:38	SELECT Name FROM Victim WHERE CrimeID IN (SELECT CrimeID FROM Crime WHERE Status = 'Closed') ...	2 row(s) returned
44	13:07:04	SELECT DISTINCT c.IncidentType FROM Crime c JOIN Victim v ON c.CrimeID = v.CrimeID WHERE v.Age > c.CrimeID	1 row(s) returned
45	13:07:57	SELECT Name FROM Victim WHERE CrimeID IN (SELECT CrimeID FROM Crime WHERE IncidentType = ...	2 row(s) returned
46	13:09:05	SELECT IncidentType, COUNT(*) AS OpenCount FROM Crime WHERE Status = 'Open' GROUP BY IncidentType	0 row(s) returned
47	13:11:54	SELECT c.* FROM Crime c JOIN Suspect s ON c.CrimeID = s.CrimeID WHERE s.Name IN (SELECT Name F...	0 row(s) returned
48	13:12:36	SELECT c.*, v.Name AS VictimName, s.Name AS SuspectName FROM Crime c LEFT JOIN Victim v ON c.CrimeID = v.CrimeID	3 row(s) returned

15. Find incidents where the suspect is older than any victim.

SELECT DISTINCT c.*

FROM Crime c

JOIN Suspect s ON c.CrimeID = s.CrimeID

WHERE s.Age > ALL (SELECT v.Age FROM Victim v WHERE v.CrimeID = c.CrimeID);

159 #15 Execute the statement under the keyboard cursor

```
160 SELECT DISTINCT c.*
161 FROM Crime c
162 JOIN Suspect s ON c.CrimeID = s.CrimeID
163 WHERE s.Age > ALL (SELECT v.Age FROM Victim v WHERE v.CrimeID = c.CrimeID);
```

164

Result Grid

CrimeID	IncidentType	IncidentDate	Location	Description	Status
1	Robbery	2023-09-15	123 Main St, Cityville	Armed robbery at a convenience store	Open
3	Theft	2023-09-10	789 Oak St, Villagetown	Shoplifting incident at a mall	Closed

Result 36 x Read Only

Output

Action Output

#	Time	Action	Message
44	13:07:04	SELECT DISTINCT c.IncidentType FROM Crime c JOIN Victim v ON c.CrimeID = v.CrimeID WHERE v.Age > ALL (SELECT v.Age FROM Victim v WHERE v.CrimeID = c.CrimeID);	1 row(s) returned
45	13:07:57	SELECT Name FROM Victim WHERE CrimeID IN (SELECT CrimeID FROM Crime WHERE IncidentType = 'Robbery');	2 row(s) returned
46	13:09:05	SELECT IncidentType, COUNT(*) AS OpenCount FROM Crime WHERE Status = 'Open' GROUP BY IncidentType;	0 row(s) returned
47	13:11:54	SELECT c.* FROM Crime c JOIN Suspect s ON c.CrimeID = s.CrimeID WHERE s.Name IN (SELECT Name FROM Suspect WHERE Name LIKE 'J%');	0 row(s) returned
48	13:12:36	SELECT c.*, v.Name AS VictimName, s.Name AS SuspectName FROM Crime c LEFT JOIN Victim v ON c.CrimeID = v.CrimeID LEFT JOIN Suspect s ON c.CrimeID = s.CrimeID;	3 row(s) returned
49	13:13:10	SELECT DISTINCT c.* FROM Crime c JOIN Suspect s ON c.CrimeID = s.CrimeID WHERE s.Age > ALL (SELECT v.Age FROM Victim v WHERE v.CrimeID = c.CrimeID);	2 row(s) returned

16. Find suspects involved in multiple incidents:

SELECT Name, COUNT(*) AS IncidentCount

FROM Suspect

GROUP BY Name

HAVING COUNT(*) > 1;

165 #16

```
166 SELECT Name, COUNT(*) AS IncidentCount
167 FROM Suspect
168 GROUP BY Name
169 HAVING COUNT(*) > 1;
```

170

Result Grid

Name	IncidentCount
------	---------------

Result 38 x Read Only

Output

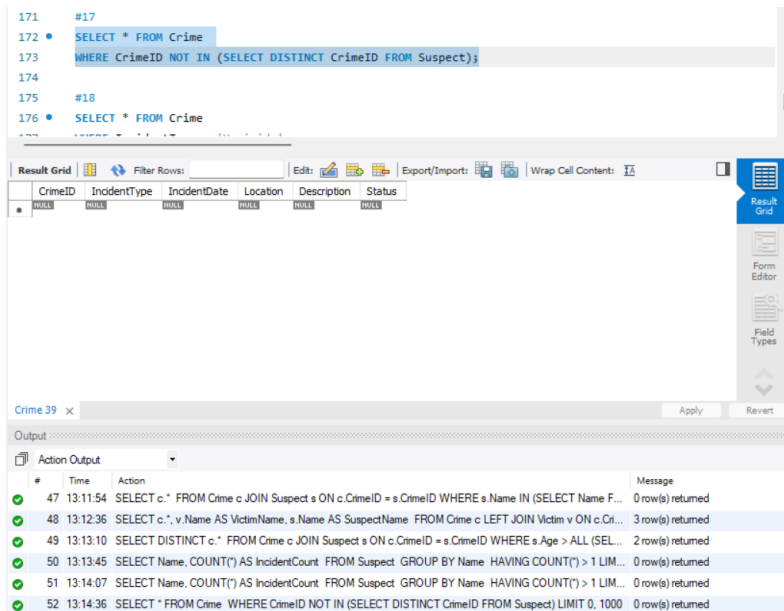
Action Output

#	Time	Action	Message
46	13:09:05	SELECT IncidentType, COUNT(*) AS OpenCount FROM Crime WHERE Status = 'Open' GROUP BY IncidentType;	0 row(s) returned
47	13:11:54	SELECT c.* FROM Crime c JOIN Suspect s ON c.CrimeID = s.CrimeID WHERE s.Name IN (SELECT Name FROM Suspect WHERE Name LIKE 'J%');	0 row(s) returned
48	13:12:36	SELECT c.*, v.Name AS VictimName, s.Name AS SuspectName FROM Crime c LEFT JOIN Victim v ON c.CrimeID = v.CrimeID LEFT JOIN Suspect s ON c.CrimeID = s.CrimeID;	3 row(s) returned
49	13:13:10	SELECT DISTINCT c.* FROM Crime c JOIN Suspect s ON c.CrimeID = s.CrimeID WHERE s.Age > ALL (SELECT v.Age FROM Victim v WHERE v.CrimeID = c.CrimeID);	2 row(s) returned
50	13:13:45	SELECT Name, COUNT(*) AS IncidentCount FROM Suspect GROUP BY Name HAVING COUNT(*) > 1 LIMIT 10;	0 row(s) returned
51	13:14:07	SELECT Name, COUNT(*) AS IncidentCount FROM Suspect GROUP BY Name HAVING COUNT(*) > 1 LIMIT 10;	0 row(s) returned

17. List incidents with no suspects involved.

SELECT * FROM Crime

WHERE CrimeID NOT IN (SELECT DISTINCT CrimeID FROM Suspect);



18. List all cases where at least one incident is of type 'Homicide' and all other incidents are of type 'Robbery'.

SELECT * FROM Crime

WHERE IncidentType = 'Homicide'

AND EXISTS (

SELECT 1 FROM Crime c2

WHERE c2.IncidentType = 'Robbery'

);

```

175 #18
176 • SELECT * FROM Crime
177 WHERE IncidentType = 'Homicide'
178 AND EXISTS (
179     SELECT 1 FROM Crime c2
180     WHERE c2.IncidentType = 'Robbery'
181 )

```

CrimeID	IncidentType	IncidentDate	Location	Description	Status
2	Homicide	2023-09-20	456 Elm St, Townsville	Investigation into a murder case	Under Inve...

Crime 40 x Apply Revert

Output

Action Output

#	Time	Action	Message
48	13:12:36	SELECT c.*, v.Name AS VictimName, s.Name AS SuspectName FROM Crime c LEFT JOIN Victim v ON c.CrimeID = v.CrimeID	3 row(s) returned
49	13:13:10	SELECT DISTINCT c.* FROM Crime c JOIN Suspect s ON c.CrimeID = s.CrimeID WHERE s.Age > ALL (SELECT s2.Age FROM Suspect s2 WHERE s2.CrimeID = c.CrimeID)	2 row(s) returned
50	13:13:45	SELECT Name, COUNT(*) AS IncidentCount FROM Suspect GROUP BY Name HAVING COUNT(*) > 1 LIMIT 10	0 row(s) returned
51	13:14:07	SELECT Name, COUNT(*) AS IncidentCount FROM Suspect GROUP BY Name HAVING COUNT(*) > 1 LIMIT 10	0 row(s) returned
52	13:14:36	SELECT * FROM Crime WHERE CrimeID NOT IN (SELECT DISTINCT CrimeID FROM Suspect) LIMIT 0, 1000	0 row(s) returned
53	13:15:34	SELECT * FROM Crime WHERE IncidentType = 'Homicide' AND EXISTS (SELECT 1 FROM Crime c2 WHERE c2.IncidentType = 'Robbery')	1 row(s) returned

19. Retrieve a list of all incidents and the associated suspects, showing suspects for each incident, or 'No Suspect' if there are none.

`SELECT c.CrimeID, c.IncidentType, COALESCE(s.Name, 'No Suspect')
AS SuspectName`

`FROM Crime c`

`LEFT JOIN Suspect s ON c.CrimeID = s.CrimeID;`

```

183 #19
184 SELECT c.CrimeID, c.IncidentType, COALESCE(s.Name, 'No Suspect') AS SuspectName
185 FROM Crime c
186 LEFT JOIN Suspect s ON c.CrimeID = s.CrimeID;
187

```

CrimeID	IncidentType	SuspectName
1	Robbery	Robber 1
2	Homicide	Unknown
3	Theft	Suspect 1

Result 41 x Read Only

Output

#	Time	Action	Message
49	13:13:10	SELECT DISTINCT c.* FROM Crime c JOIN Suspect s ON c.CrimeID = s.CrimeID WHERE s.Age > ALL (SEL...	2 row(s) returned
50	13:13:45	SELECT Name, COUNT(*) AS IncidentCount FROM Suspect GROUP BY Name HAVING COUNT(*) > 1 LIM...	0 row(s) returned
51	13:14:07	SELECT Name, COUNT(*) AS IncidentCount FROM Suspect GROUP BY Name HAVING COUNT(*) > 1 LIM...	0 row(s) returned
52	13:14:36	SELECT * FROM Crime WHERE CrimeID NOT IN (SELECT DISTINCT CrimeID FROM Suspect) LIMIT 0, 1000	0 row(s) returned
53	13:15:34	SELECT * FROM Crime WHERE IncidentType = 'Homicide' AND EXISTS (SELECT 1 FROM Crime c2 ...	1 row(s) returned
54	13:16:15	SELECT c.CrimeID, c.IncidentType, COALESCE(s.Name, 'No Suspect') AS SuspectName FROM Crime c LEF...	3 row(s) returned

20. List all suspects who have been involved in incidents with incident types 'Robbery' or 'Assault'

SELECT s.*

FROM Suspect s

JOIN Crime c ON s.CrimeID = c.CrimeID

WHERE c.IncidentType IN ('Robbery', 'Assault');

```

188 #20
189 SELECT s.*
190 FROM Suspect s
191 JOIN Crime c ON s.CrimeID = c.CrimeID
192 WHERE c.IncidentType IN ('Robbery', 'Assault');

```

SuspectID	CrimeID	Name	Description	CriminalHistory	Age
1	1	Robber 1	Armed and masked robber	Previous robbery convictions	40

Result 42 x Read Only

Output

#	Time	Action	Message
50	13:13:45	SELECT Name, COUNT(*) AS IncidentCount FROM Suspect GROUP BY Name HAVING COUNT(*) > 1 LIM...	0 row(s) returned
51	13:14:07	SELECT Name, COUNT(*) AS IncidentCount FROM Suspect GROUP BY Name HAVING COUNT(*) > 1 LIM...	0 row(s) returned
52	13:14:36	SELECT * FROM Crime WHERE CrimeID NOT IN (SELECT DISTINCT CrimeID FROM Suspect) LIMIT 0, 1000	0 row(s) returned
53	13:15:34	SELECT * FROM Crime WHERE IncidentType = 'Homicide' AND EXISTS (SELECT 1 FROM Crime c2 ...	1 row(s) returned
54	13:16:15	SELECT c.CrimeID, c.IncidentType, COALESCE(s.Name, 'No Suspect') AS SuspectName FROM Crime c LEF...	3 row(s) returned
55	13:16:51	SELECT s.* FROM Suspect s JOIN Crime c ON s.CrimeID = c.CrimeID WHERE c.IncidentType IN ('Robbery', ...	1 row(s) returned